

Course Announcements

Due this Sunday: D8, Q9, A4, EDA Checkpoint

Notes:

- Proposal/Checkpoint Scores Posted & Feedback Available
 - Please leave all grading issues open
- Please complete your SETs (we'll also have a post-course survey)
- Caught up on Ed posts yesterday - so if something not answered, make sure it's unresolved

Geospatial Analysis

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How a coastline 100 million years ago
influences modern election results in Alabama

Cretaceous Sediments



Fertile Blackland
Prairie Soil



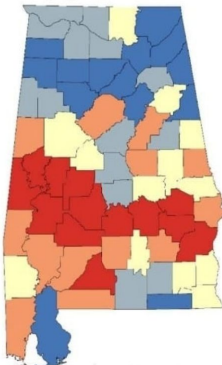
Average Farm Size, 1997



Slave Population, 1860



Black population, 2010



-Starkey Comics

Election Results, 2020



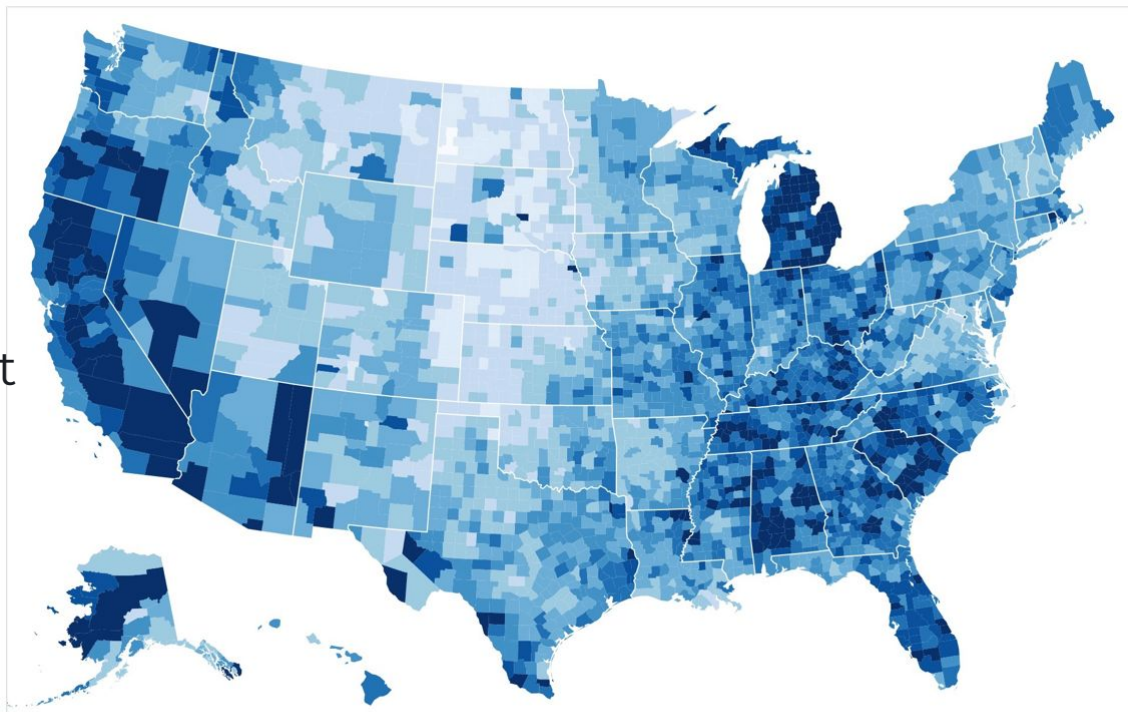
Why Geospatial Analysis?

“Everything is related to everything else, but near things are more related than distant things.” -Tobler 1979

“...the purpose of geographic inquiry is to examine relationships between geographic features collectively and to use the relationships to describe the real-world phenomena that map features represent” -Clarke 2001

Visualizing Geospatial Data

Unemployment
rate by county
(August 2016)



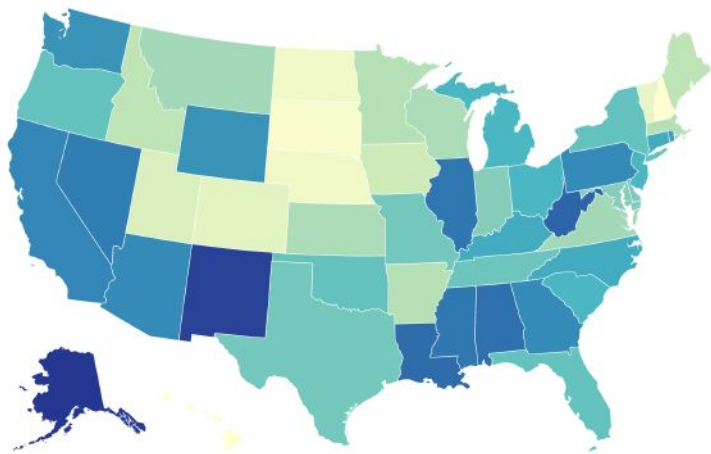
This choropleth encodes unemployment rates from 2008 with a [quantize scale](#) ranging from 0 to 15%. A [threshold scale](#) is a useful alternative for coloring arbitrary ranges.

[Open in a new window.](#)

Choropleth maps are useful for visualizing *clear regional patterns* in the data

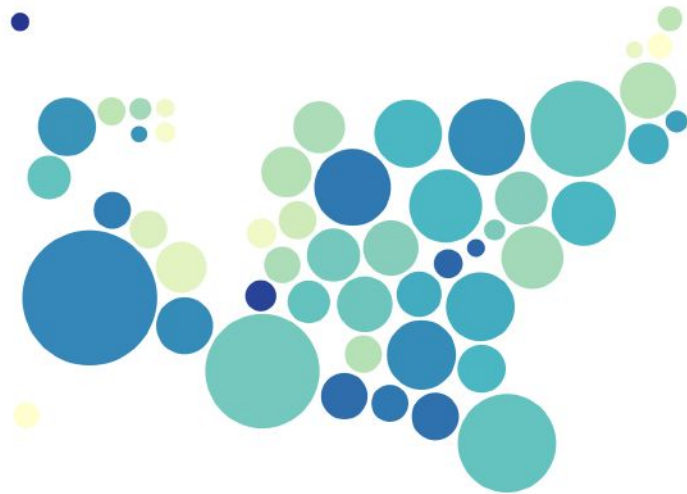
Cartograms should be considered when displaying how many people were affected

NOT IDEAL



Choropleths answer "How much area was affected?"

BETTER



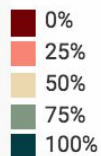
Cartograms answer "How many people were affected?"

Share of individuals using the internet, 2015

Share of individuals using the internet, measured as the percentage of the population. Internet users are individuals who have used the Internet (from any location) in the last 3 months. The Internet can be used via a computer, mobile phone, personal digital assistant, games machine, digital TV etc.

- ☐ **A** Thought about it and have a thought
- ☐ **B** Thought about it and have no thought
- ☐ **C** I'm confused

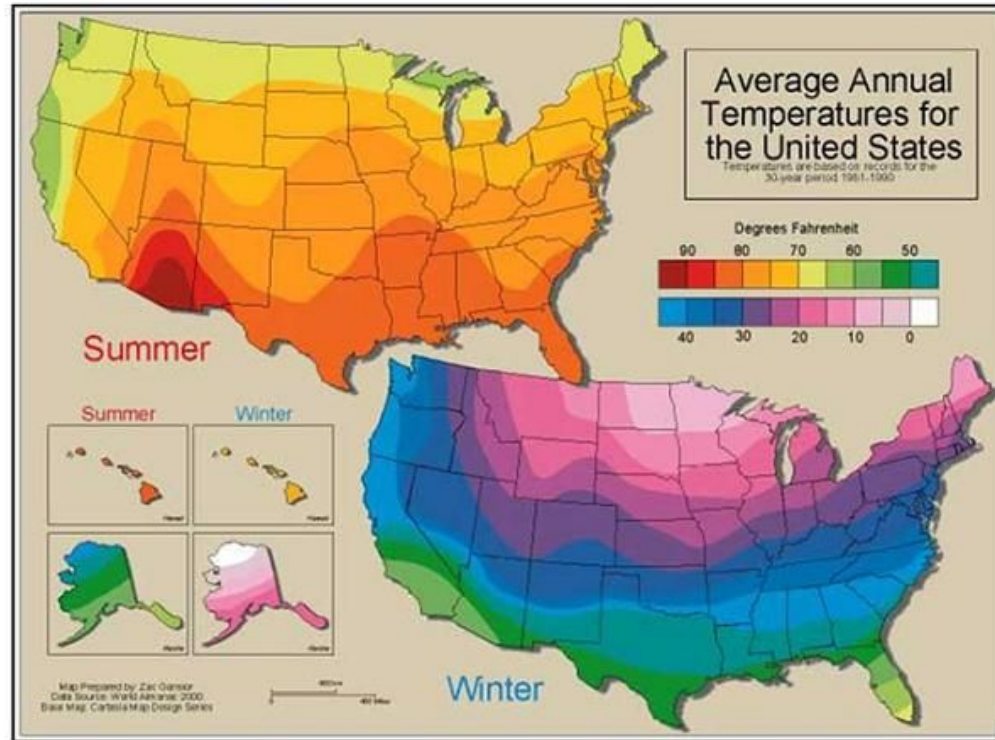
Share of people using the internet



Source: [Our World in Data](#) • [Get the data](#)

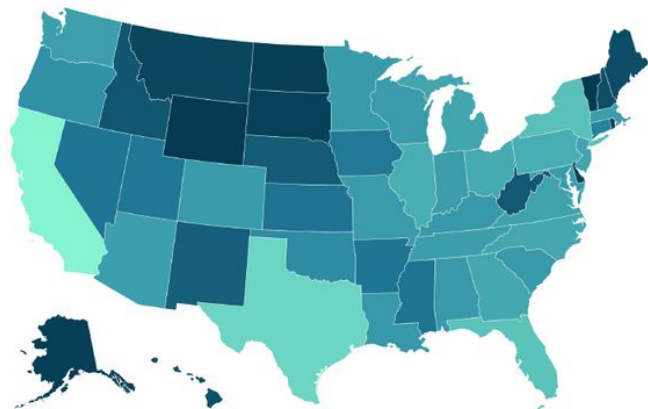


Isarithmic maps demonstrate smooth, continuous phenomena (temperature, elevation, rainfall, etc.)



Use light colors for low values. Dark colors for high values.

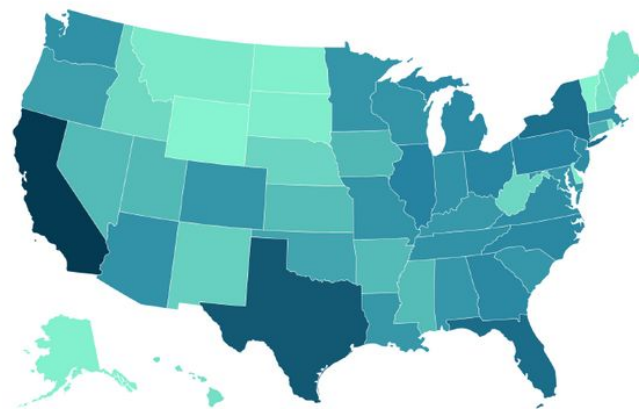
NOT IDEAL



LOW POPULATION HIGH



BETTER



LOW POPULATION HIGH



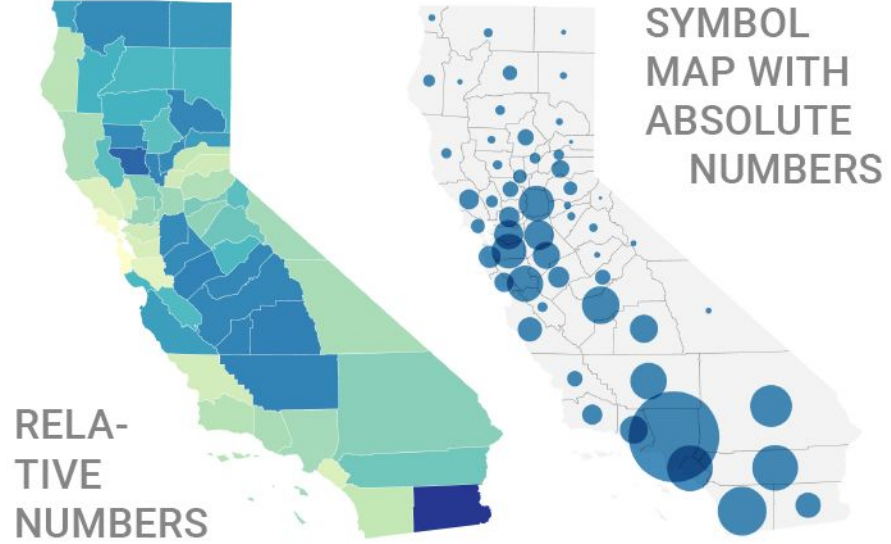
Choropleth maps are useful for visualizing *clear regional patterns* in the data

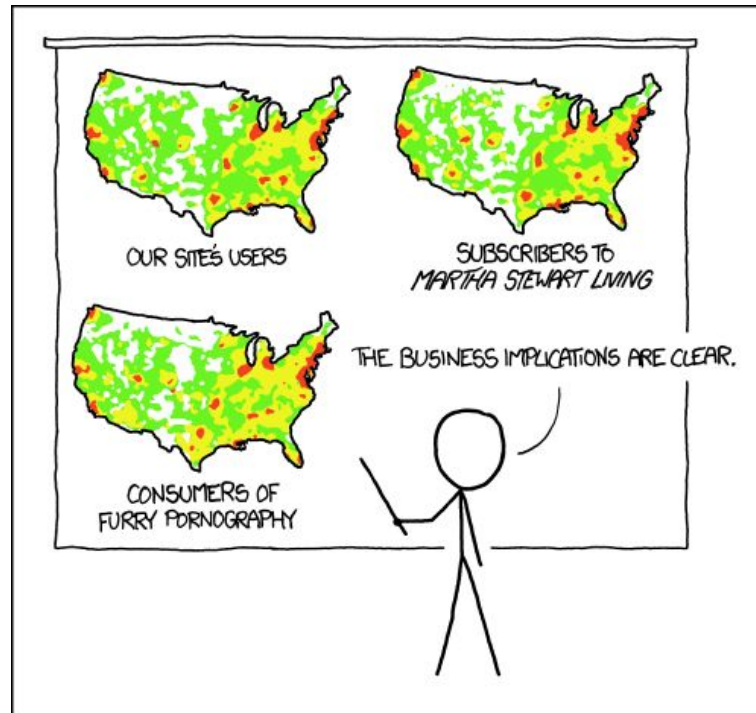
Choropleth should display relative differences, *not* absolute numbers

NOT IDEAL



BETTER



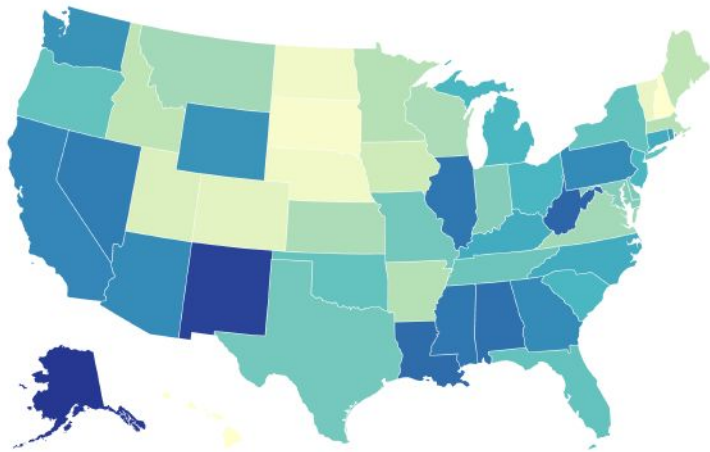


PET PEEVE #208:
GEOGRAPHIC PROFILE MAPS WHICH ARE
BASICALLY JUST POPULATION MAPS

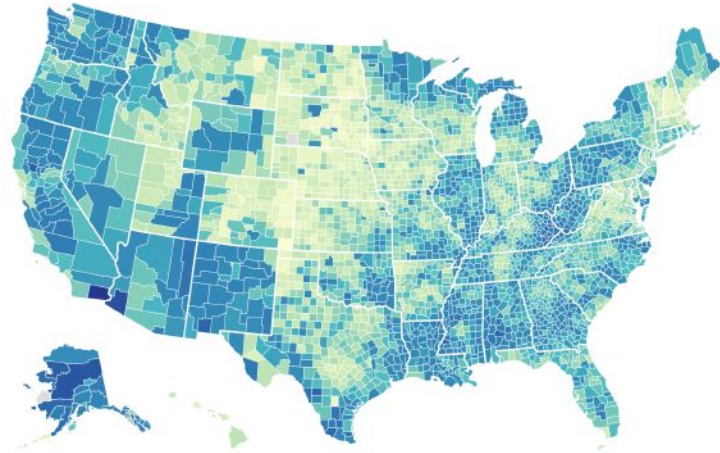
Choropleth maps can be misleading

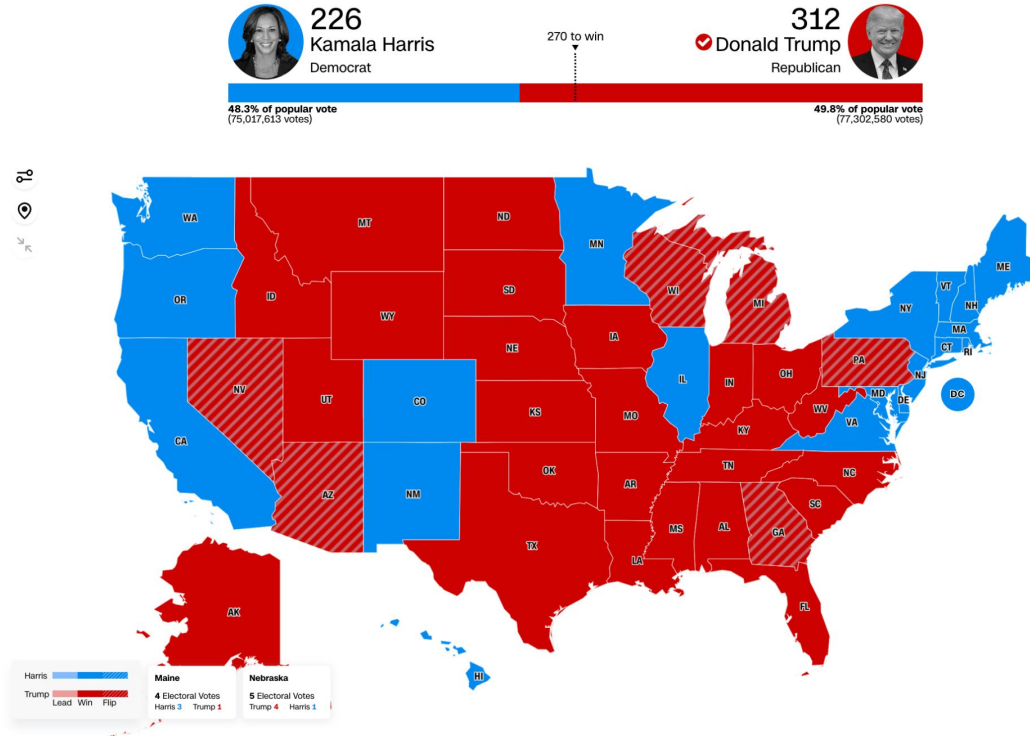
Consider using the smallest unit possible (but there are exceptions!)

NOT IDEAL



BETTER

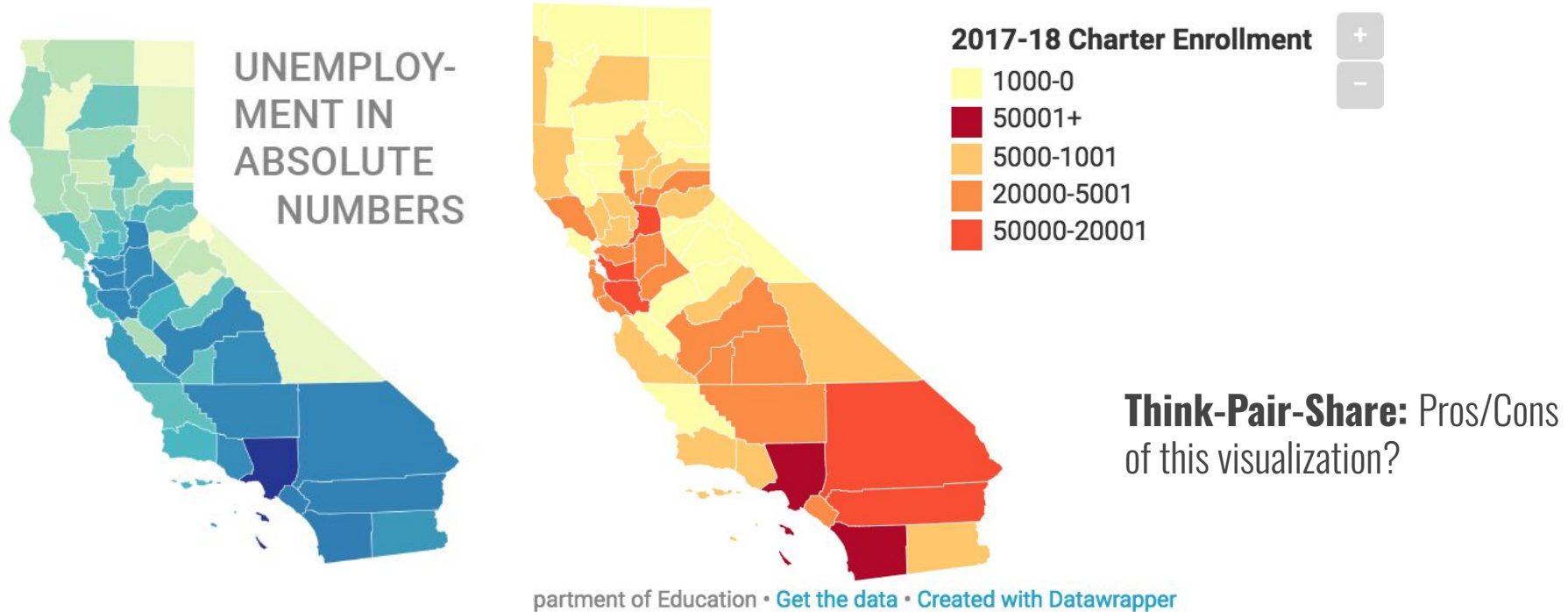




Sometimes summarizing at a bigger unit ok...

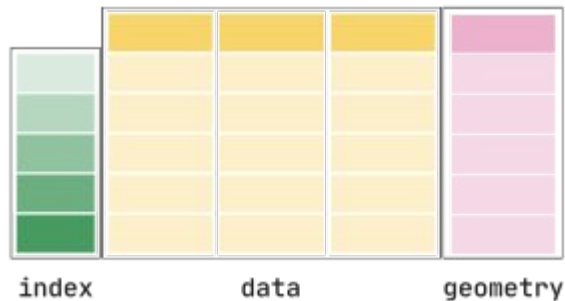
Map: Where Are Students Attending Charter Schools?

The majority of California's charter school student population is concentrated in Los Angeles, San Diego and Bay Area counties. Hover through the counties on each map for more information on their



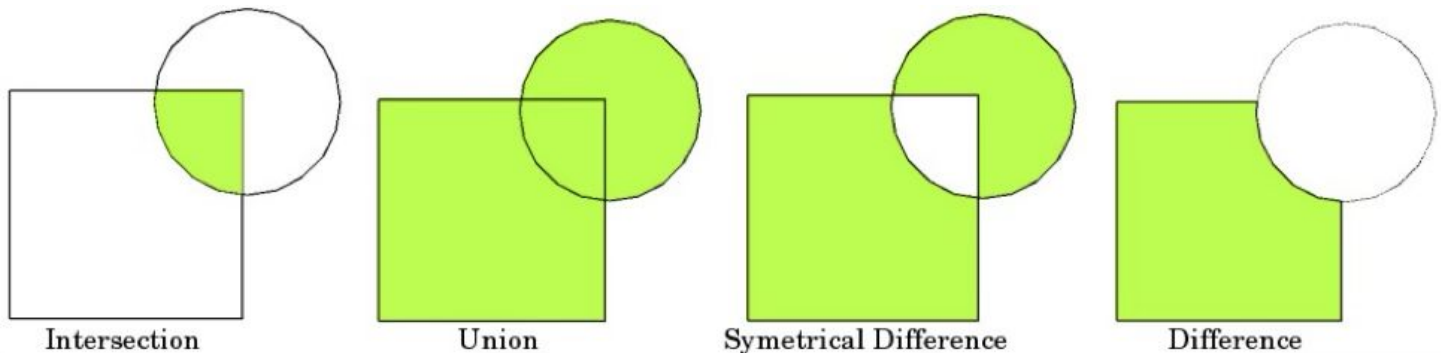
geopandas: Working with maps/geospatial data in Python

- Extends pandas, adds support for geospatial data
- Uses shapely library to represent vector geometry (e.g., POINT, LINE, POLYGON)
- Writes/reads shapefiles, GeoJSON and other formats
- Uses Coordinate Reference Systems (CRS) to represent how data aligns with the real world
- Different types of CRS: Geographic Coordinate Systems and Projected Coordinate Systems.



geopandas skills

- Overlay spatial datasets: e.g. calculate what regions of Chicago are within 1km of a grocery store
- Join dataframes on shape information
- Aggregate across different subregions to get measurements at a more granular regional level



Geospatial/GIS Resources:

geopandas tutorials:

- https://geopandas.org/en/stable/getting_started/introduction.html
- https://geopandas.org/en/stable/docs/user_guide/mapping.html
- https://geopandas.org/en/stable/docs/user_guide/set_operations.html
- <https://geopandas.org/en/stable/gallery/index.html>

Textbook: <https://mgimond.github.io/Spatial/introGIS.html>

Software: ArcGIS (UCSD [link](#))

Spatial Statistics : The Why

Spatial Statistics

The statistical techniques we've discussed so far don't work well when considering spatial distributions...

Spatial Statistics

The statistical techniques we've discussed so far don't work well when considering spatial distributions...

...which means we have a chance to take a look at data and the relationship between the data in new and interesting ways
(distance, adjacency, interaction, and neighbor)

Spatial data violate conventional statistics:

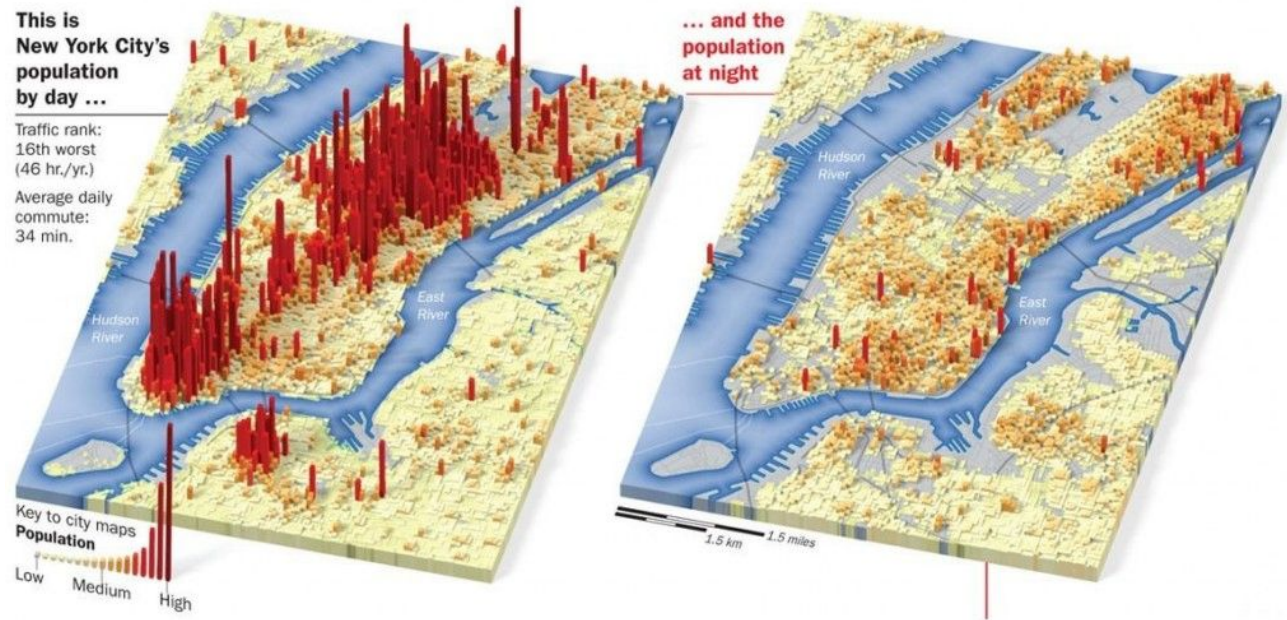
Violations of conventional statistics:

- Spatial autocorrelation
- Modifiable areal unit problem (MAUP)
- Edge effects (Boundary problem)
- Ecology fallacy
- Nonuniformity of space

Spatial Autocorrelation

Data from locations near one another in space are more likely to be similar than data from locations remote from one another:

- Housing market
- Elevation change
- Temperature



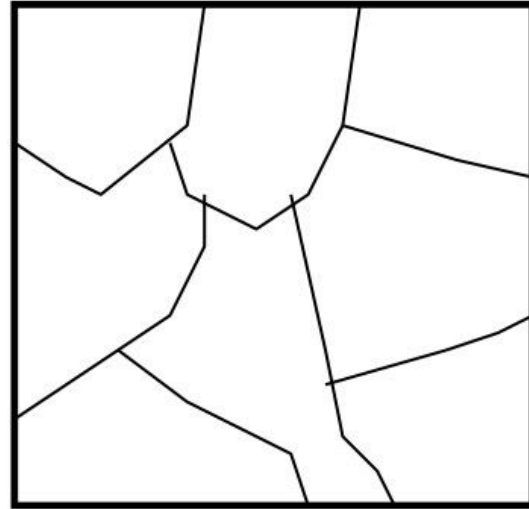
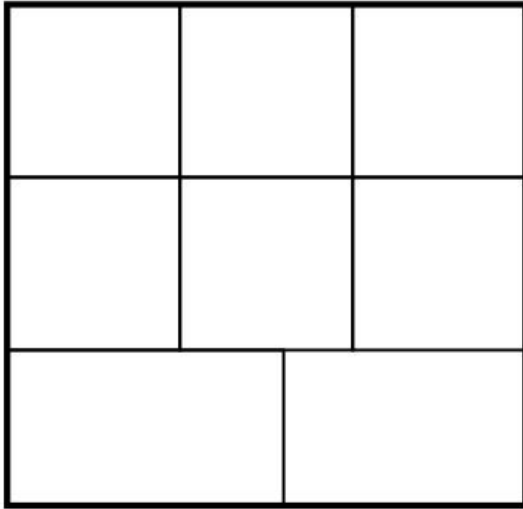
Modifiable Areal Unit Problem (MAUP)

The aggregation units used are arbitrary with respect to the phenomena under investigation, yet the aggregation units used will affect statistics determined on the basis of data reported in this way.

If the spatial units in a particular study were specified differently, we might observe very different patterns and relationships.

Modifiable Areal Unit Problem (MAUP)

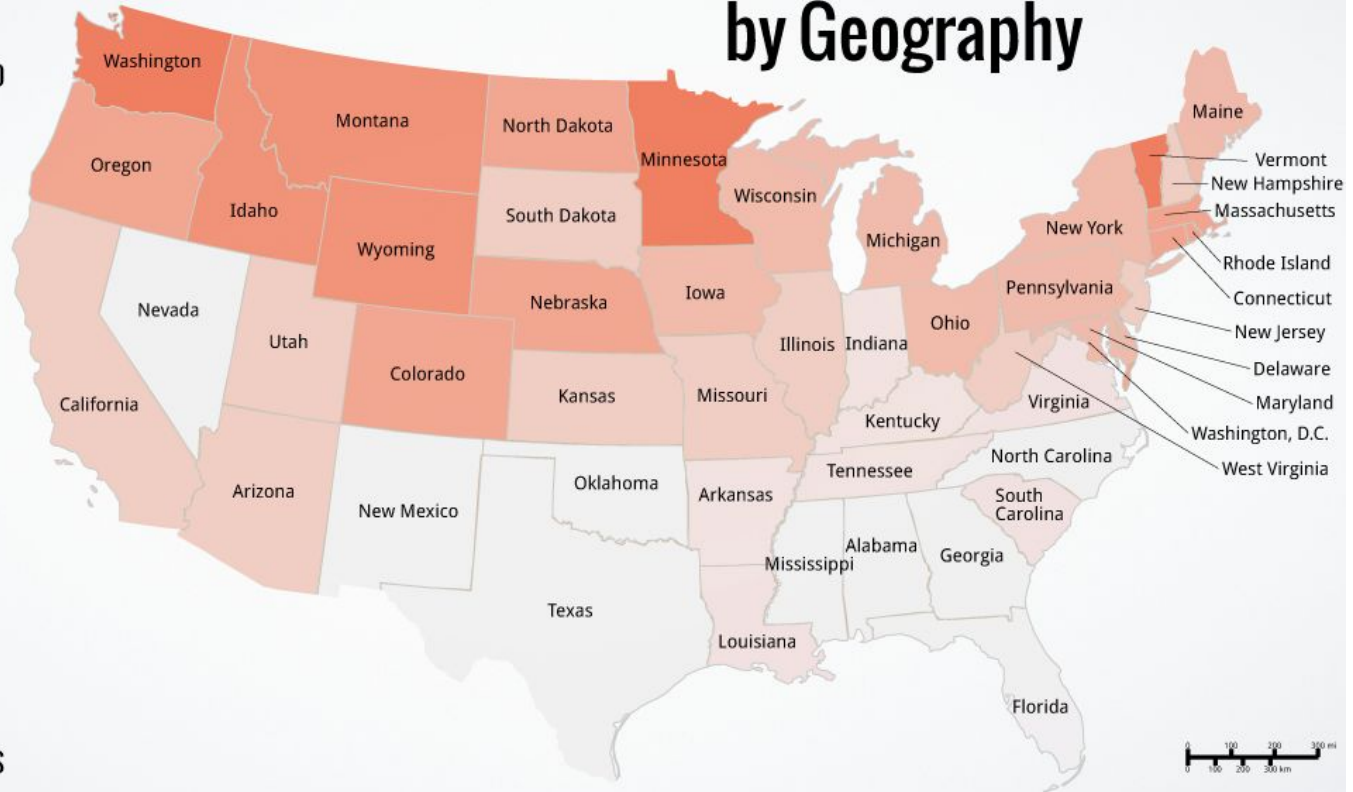
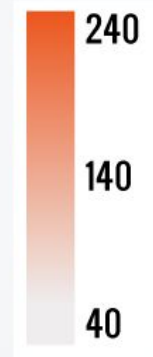
modifiable area: Units are arbitrary defined and different organization of the units may create different analytical results.



Multiple Sclerosis by Geography

CASE-CONTROL RATIO OF MS

A higher ratio indicates
greater prevalence



LEARN MORE AT
WWW.INVW.ORG/MS

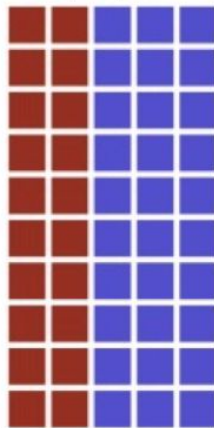
PRODUCED BY: JASON ALCORN/INVESTIGATEWEST SOURCE: BERETICH AND BERETICH (2009)

For example...gerrymandering

Gerrymandering, explained

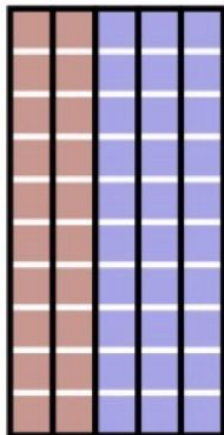
Three different ways to divide 50 people into five districts

50
people



60% blue,
40% red

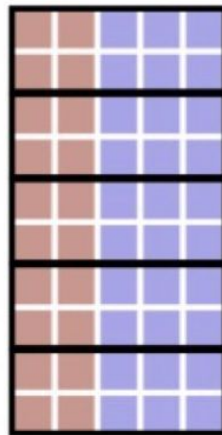
1. Perfect
representation



3 blue districts,
2 red districts

BLUE WINS

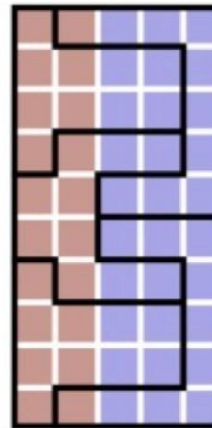
2. Compact,
but unfair



5 blue districts,
0 red districts

BLUE WINS

3. Neither compact
nor fair



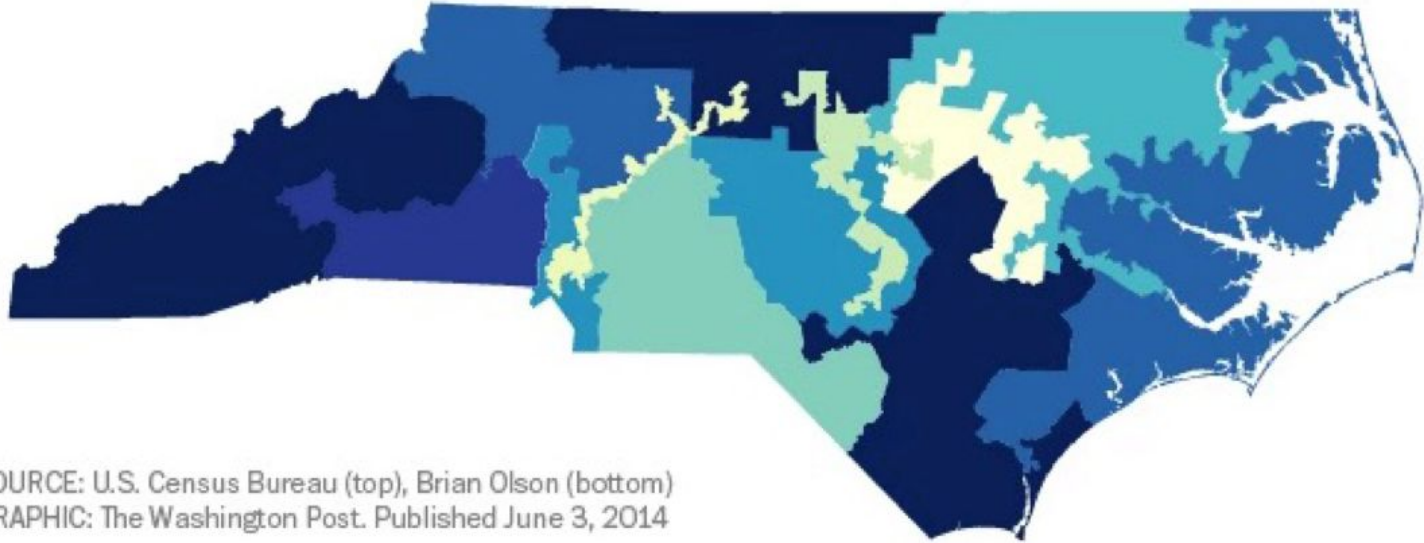
2 blue districts,
3 red districts

RED WINS

For example...gerrymandering

North Carolina

CURRENT CONGRESSIONAL DISTRICTS



SOURCE: U.S. Census Bureau (top), Brian Olson (bottom)
GRAPHIC: The Washington Post. Published June 3, 2014

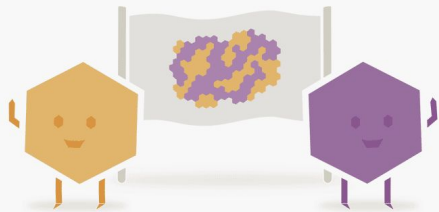
For example...gerrymandering

North Carolina

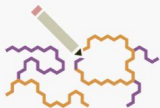
DISTRICTS REDRAWN TO OPTIMIZE COMPACTNESS



SOURCE: U.S. Census Bureau (top), Brian Olson (bottom)
GRAPHIC: The Washington Post. Published June 3, 2014



Welcome to Hexapolis



Every 10 years, Hexapolis redraws its congressional district lines — just like the United States does. But Hexapolis is a simpler place.



Lawmakers in either the **Purple Party** or **Yellow Party** control redistricting. To increase their advantage in upcoming elections, they have been known to gerrymander egregiously — even if it means leaving some voters disenfranchised.



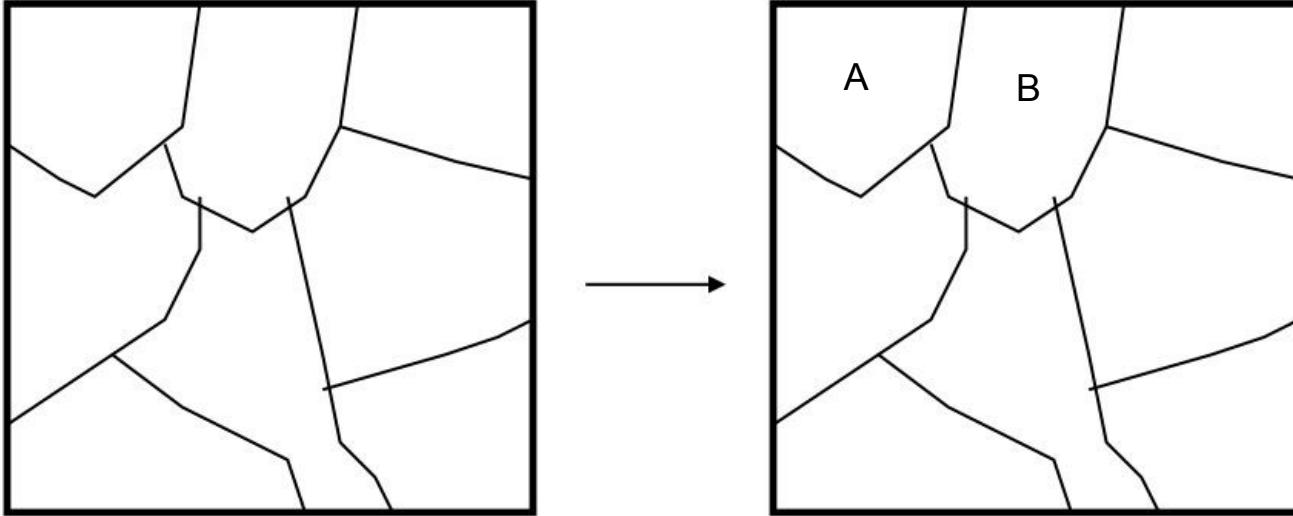
Hexapolis has nine districts. Even though a majority of voters favor the Purple Party, that does not mean that the Yellow Party can't shift the state's partisan tilt.

Gerrymandering Redistricting Game:

<https://www.nytimes.com/interactive/2022/01/27/us/politics/congressional-gerrymandering-redistricting-game-2022.html>

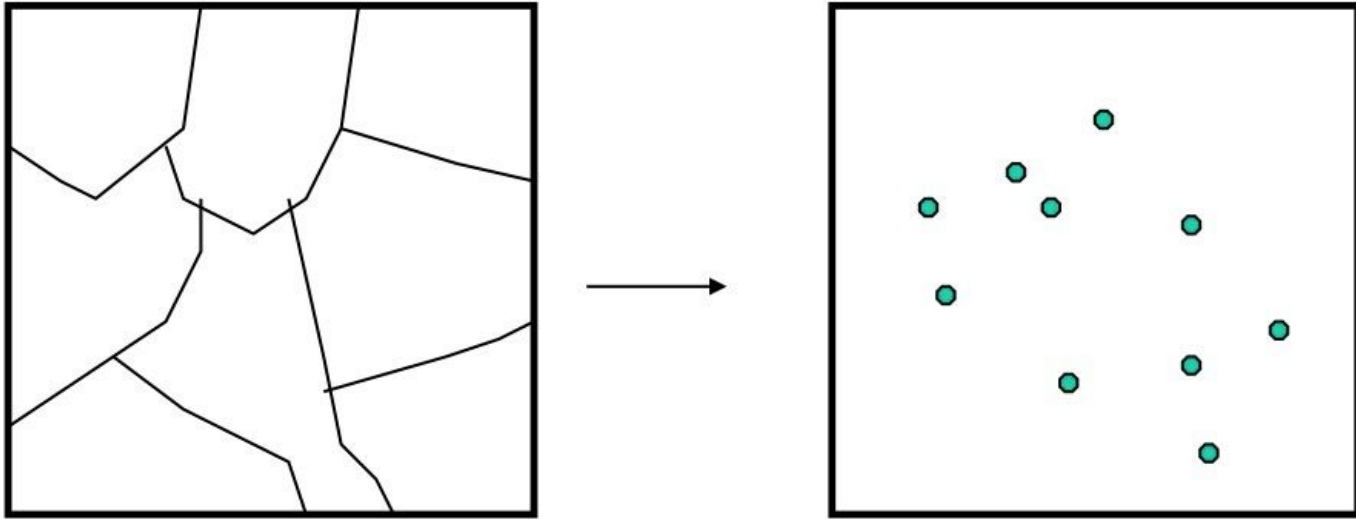
Edge Effects (The Boundary Problem)

Analyzing A vs B ignores similarities
between the two based on their
shared boundary



Ecological Fallacy

The Ecological Fallacy is a situation that can occur when a researcher or analyst makes an inference about an individual based on aggregate data for a group.



Ecological Fallacy

Example: we might observe a *strong relationship between income and crime at the county level*, with lower-income areas being associated with higher crime rate.

Conclusion:

- Lower-income persons are more likely to commit crime
- Individuals from lower-income areas are more likely to commit crime
- Lower-income counties tend to experience higher crime rates **The only valid conclusion!**

Ecological Fallacy

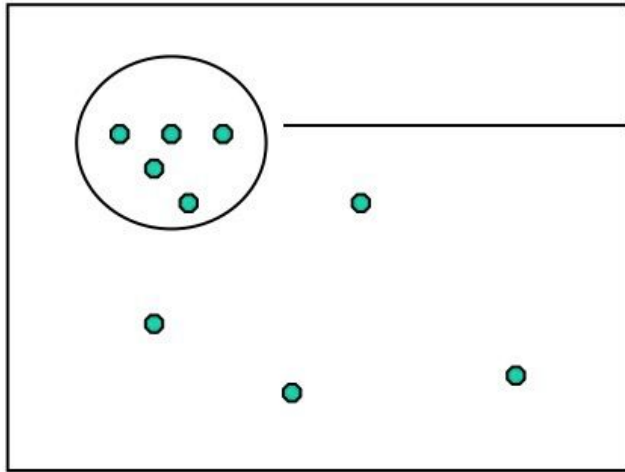
Issues:

Inferences drawn about associations between the characteristics of an aggregate population and the characteristics of sub-units within the population are wrong. That is: *results from aggregated data (e.g. counties) cannot be applied to individual people*

What should we do?

Be aware of the process of aggregating or disaggregating data may conceal the variations that are not visible at the larger aggregate level

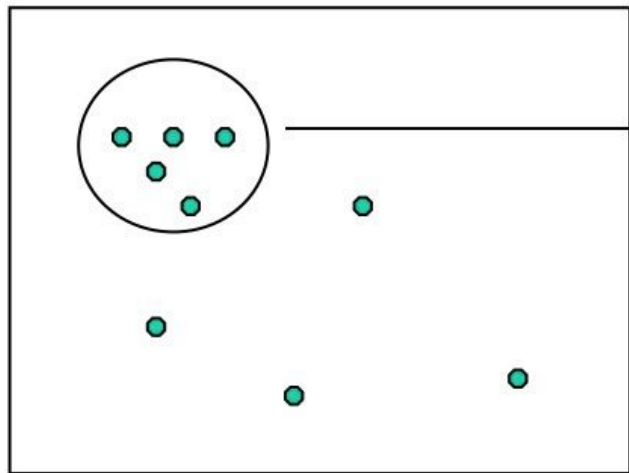
Nonuniformity



Area with high crime rates?

Crime locations

Nonuniformity



Area with high crime rates?

Conclusion: Bank robberies are clustered
....but only because banks are clustered!

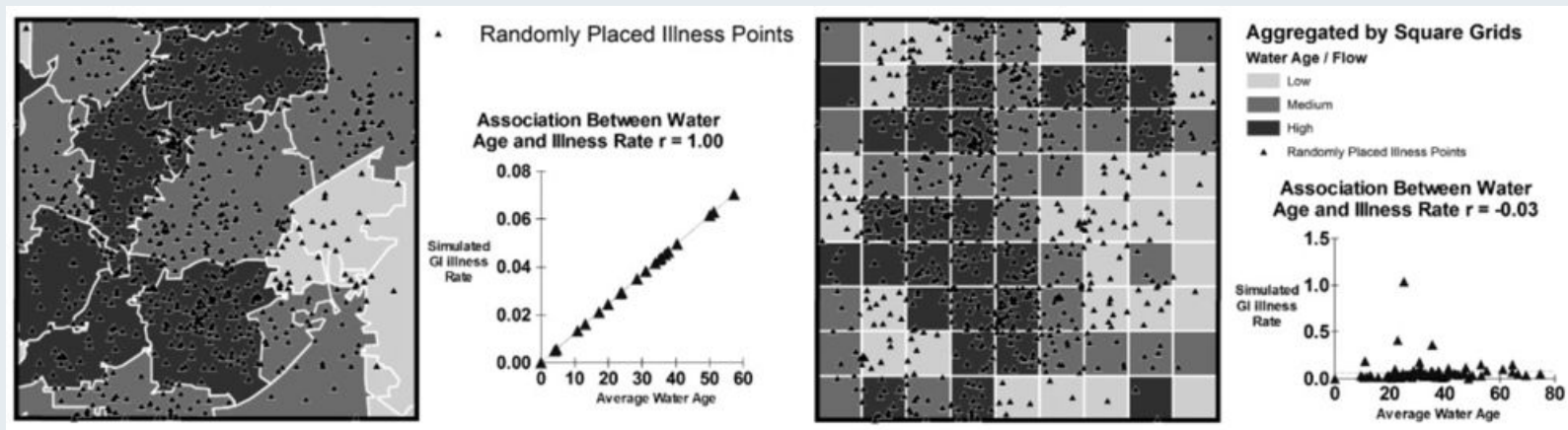
Crime locations

Checking our understanding

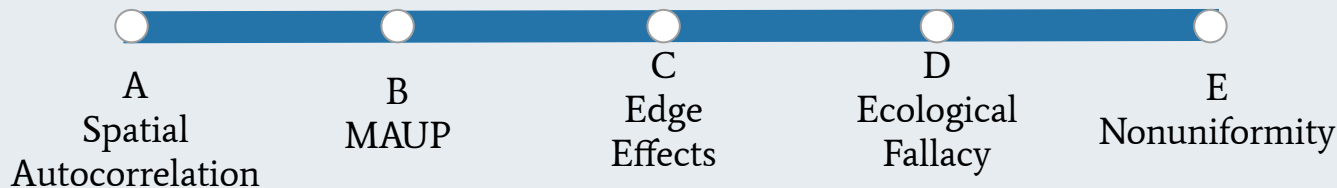
Answer all three questions in the form



Spatial Statistics



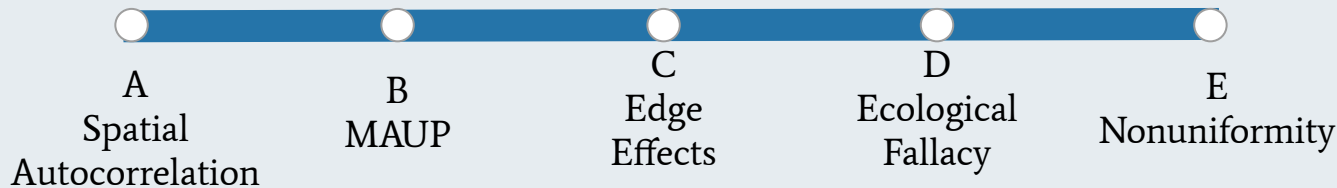
What explains what's going on here?



Spatial Statistics

In Baltimore City, police spend more time in a few neighborhoods. Crime rates are higher in those neighborhoods.

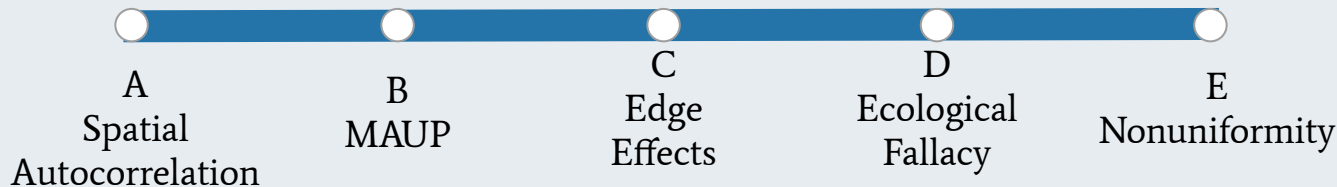
What explains what's going on here?



Spatial Statistics

A Trader Joe's just opened in a new neighborhood.
Nearby homes are now worth more money.

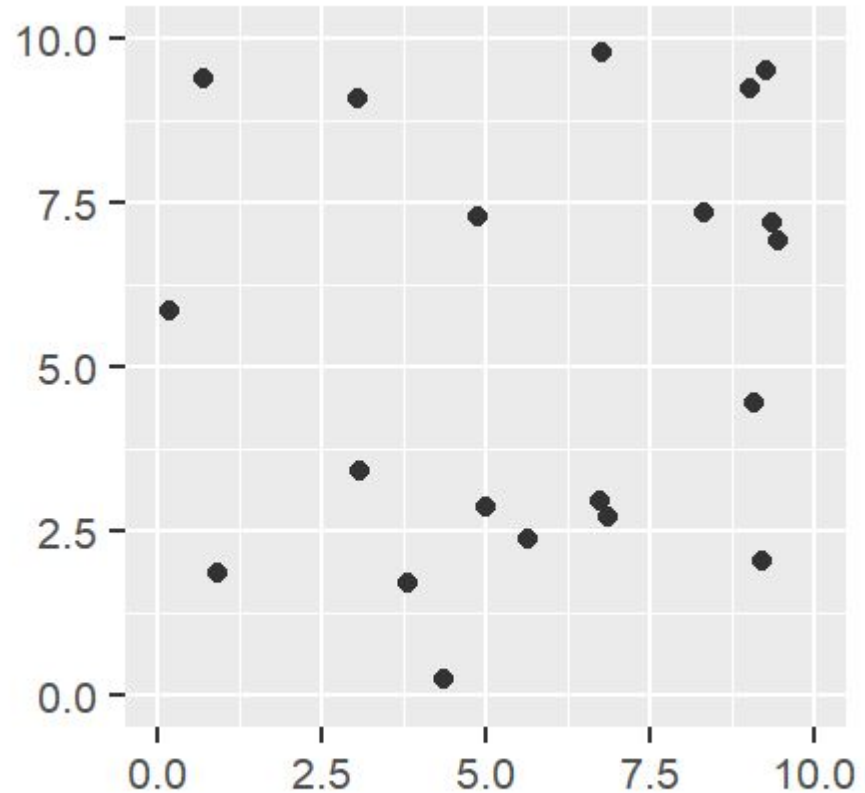
What explains what's going on here?



Spatial Statistics : The Basics

Global Point Density

the ratio of observed number of points to the study region's surface area



Quadrat Density (local)

Surface is divided and then point density is calculated within quadrat

Note: quadrat number and shape will affect measurement estimate.

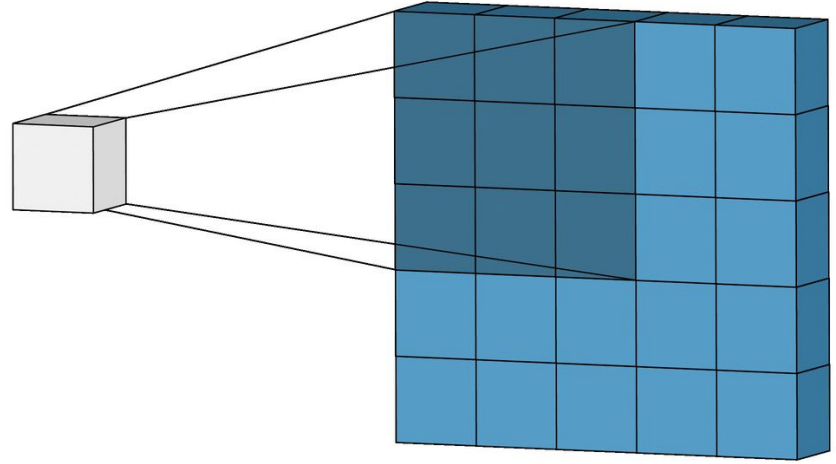
Suffers from MAUP.



Kernel Density (local)

Point density is calculated within sliding windows (window size = kernel)

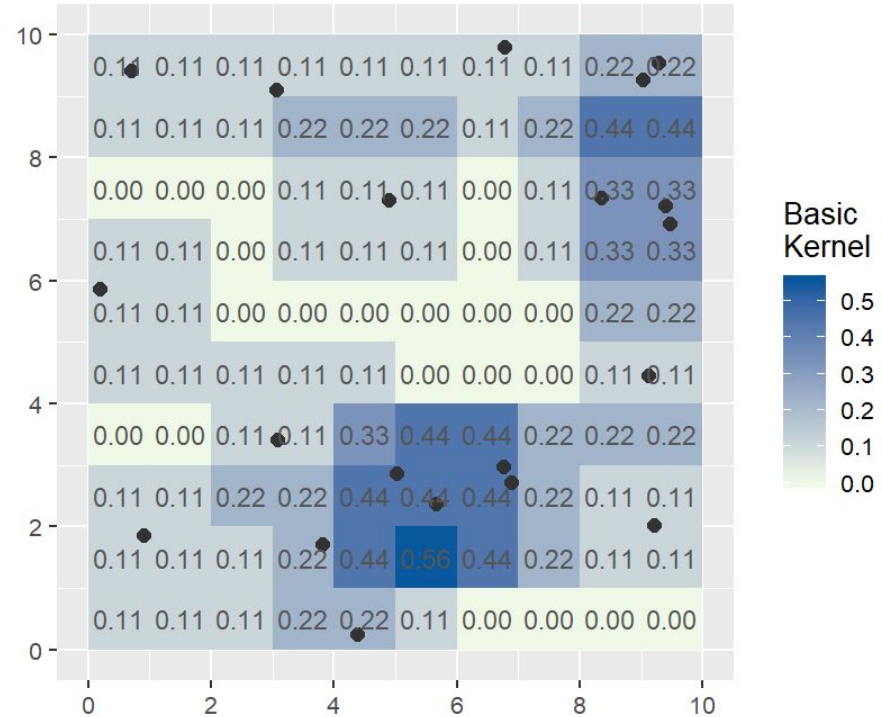
Note: kernel will affect measurement estimate, but this is less susceptible to MAUP.



Kernel Density (local)

Point density is calculated within sliding windows (window size = kernel)

Note: kernel will affect measurement estimate, but this is less susceptible to MAUP.



Modeling these data: Poisson Point Process

(Density-based Methods - - how the points are distributed relative to the study space)

$$\lambda(i) = e^{\alpha + \beta Z(i)}$$

$\lambda(i)$ is the modeled intensity at location i

e^{α} is the base intensity when the covariate is *zero*

e^{β} is the multiplier by which the intensity increases (or decreases) for each 1 unit increase in the covariate

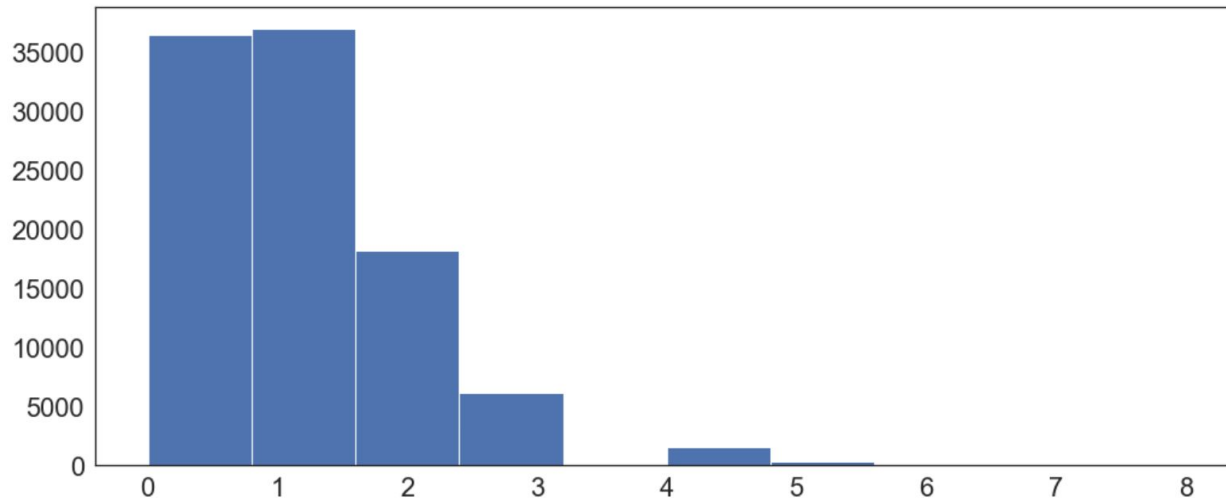
Poisson Distribution

The Poisson Distribution models events in fixed intervals of time, given a known average rate (and independent occurrences).

In [55]:

Slide Type Fragment

```
dat = poisson.rvs(mu=1, size=100000)
plt.hist(dat);
```

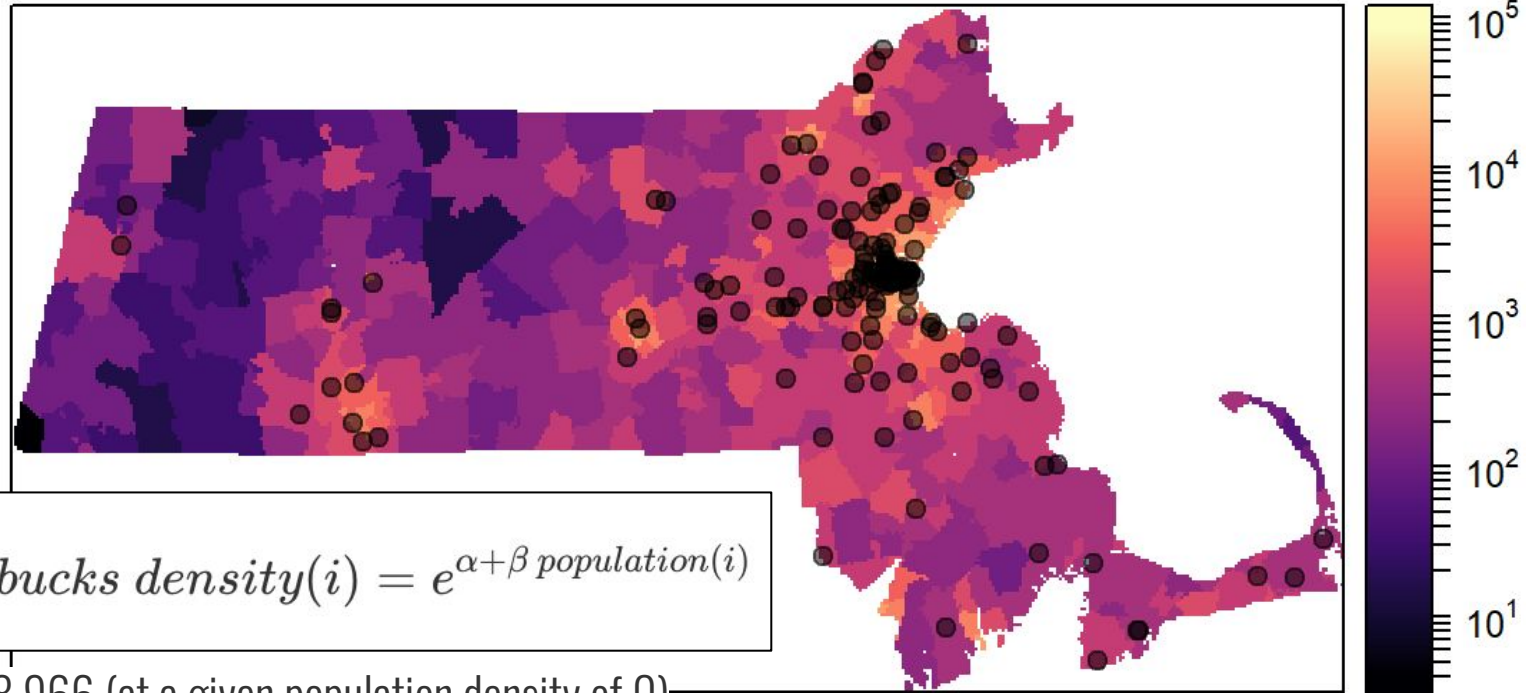


Slide Type Fragment

The **number of visitors a fast food drive-through gets each minute** follows a Poisson distribution. In this case, maybe the average is 3, but there's some variability around that number.

A Poisson distribution can help calculate the probability of various events related to customers going through the drive-through at a restaurant. It will predict lulls (0 customers) and flurry of activity (5+ customers), allowing staff to plan and schedule more precisely.

Location of Starbucks relative to population density in MA



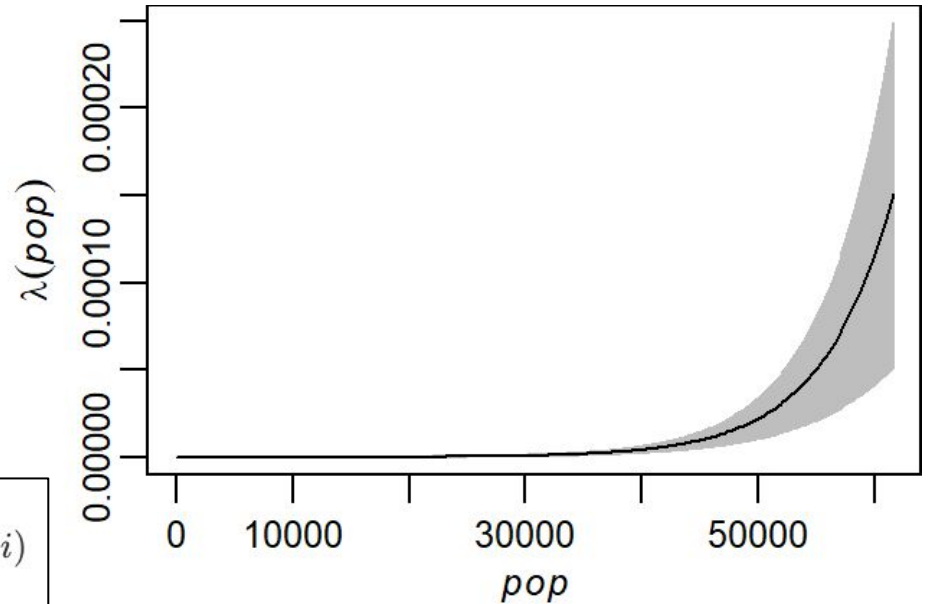
$$\text{Starbucks density}(i) = e^{\alpha + \beta \text{ population}(i)}$$

$\alpha = -18.966$ (at a given population density of 0)

$e^{-18.966} = 5.80 \times 10^{-9}$ cafes per square meter

$\beta = 0.00017$; $e^{0.00017}$ or 1.00017

Location of Starbucks relative to population density in MA



$$\text{Starbucks density}(i) = e^{\alpha + \beta \text{ population}(i)}$$

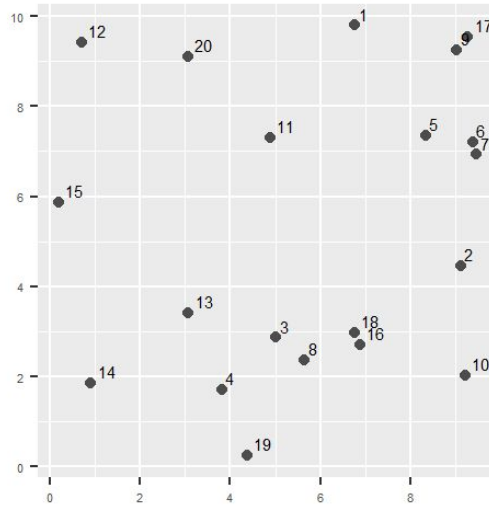
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Modeling these data: Average Nearest Neighbor

(Distance-based Methods - how the points are distributed relative to one another)



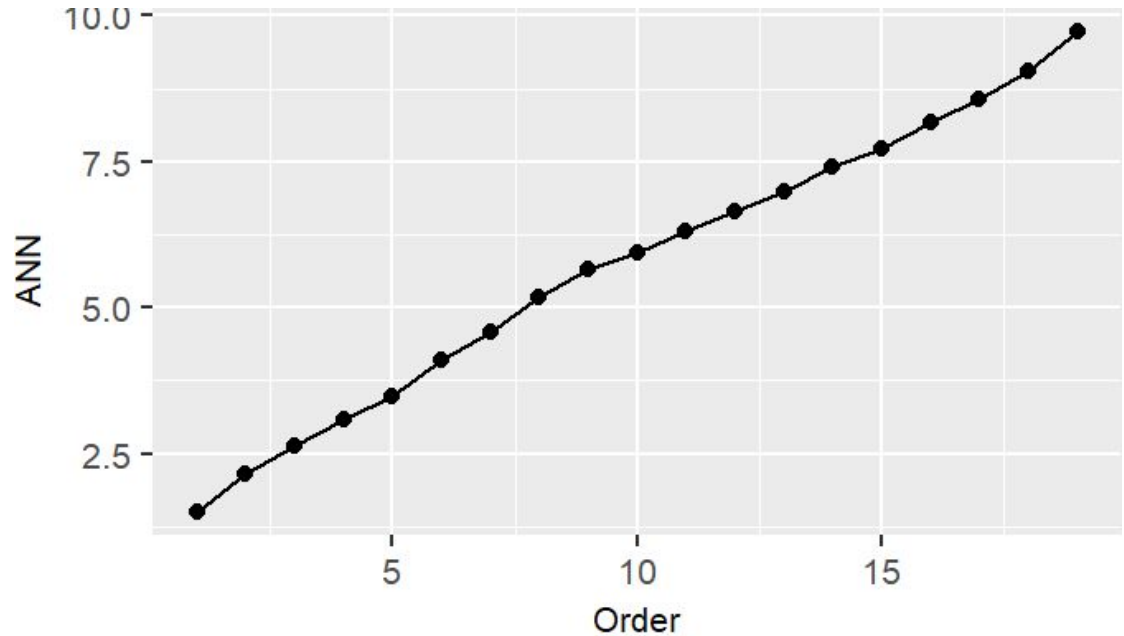
From	To	Distance	From	To	Distance
1	9	2.32	11	20	2.55
2	10	2.43	12	20	2.39
3	8	0.81	13	4	1.85
4	19	1.56	14	13	2.67
5	6	1.05	15	12	3.58
6	7	0.3	16	18	0.29
7	6	0.3	17	9	0.37
8	3	0.81	18	16	0.29
9	17	0.37	19	4	1.56
10	2	2.43	20	12	2.39

ANN = 1.52 units

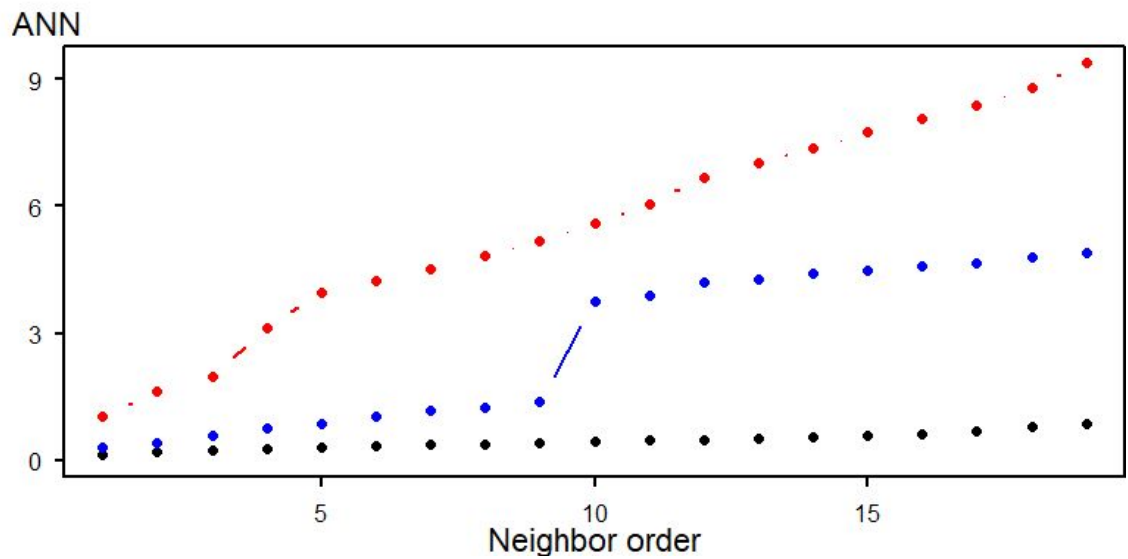
Modeling these data: Average Nearest Neighbor

(Distance-based Methods - how the points are distributed relative to one another)

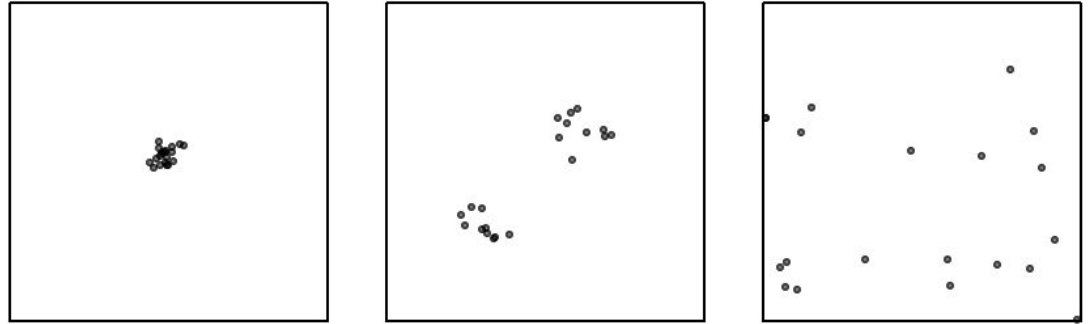
plot the ANN values for different order neighbors, that is for the first closest point, then the second closest point, and so forth.



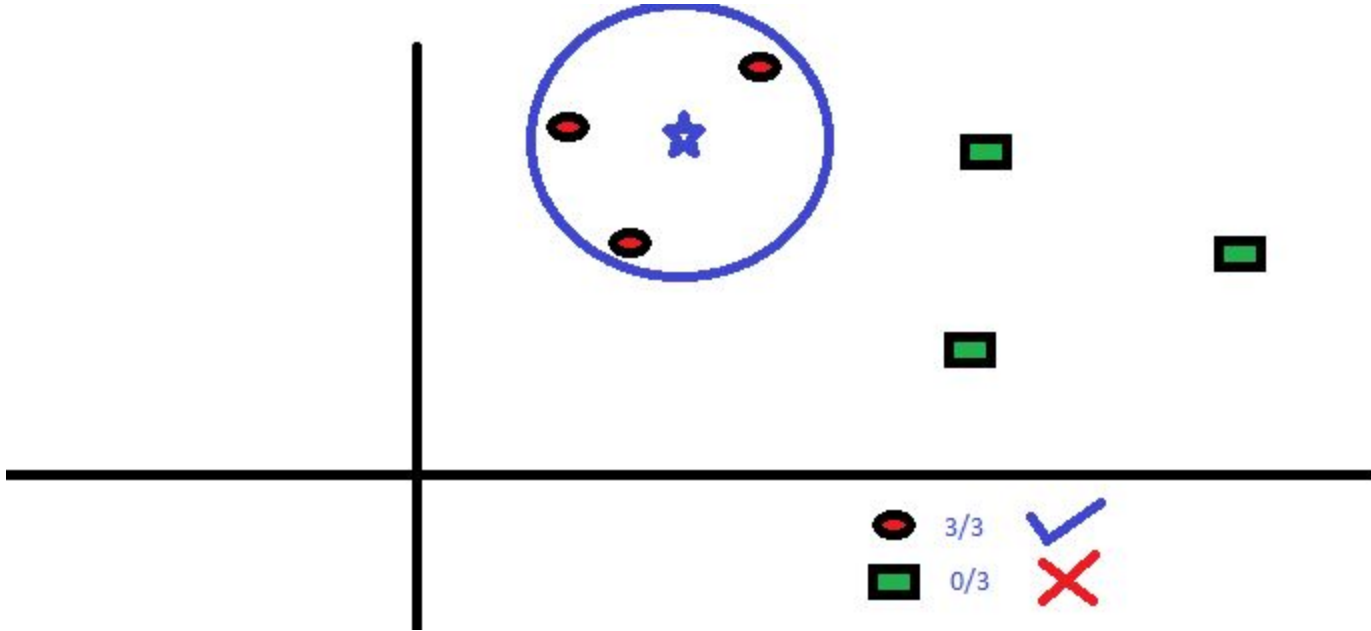
ANN vs neighbor order offers insight into underlying spatial relationship



Note: study space definition affects this measure



KNN: To which class does the blue star belong?



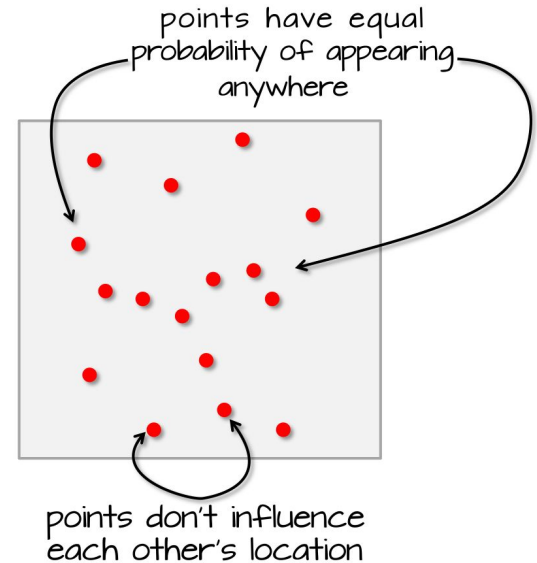
Hypothesis Testing: CSR/IPR

(Distance-based Methods - how the points are distributed relative to one another)

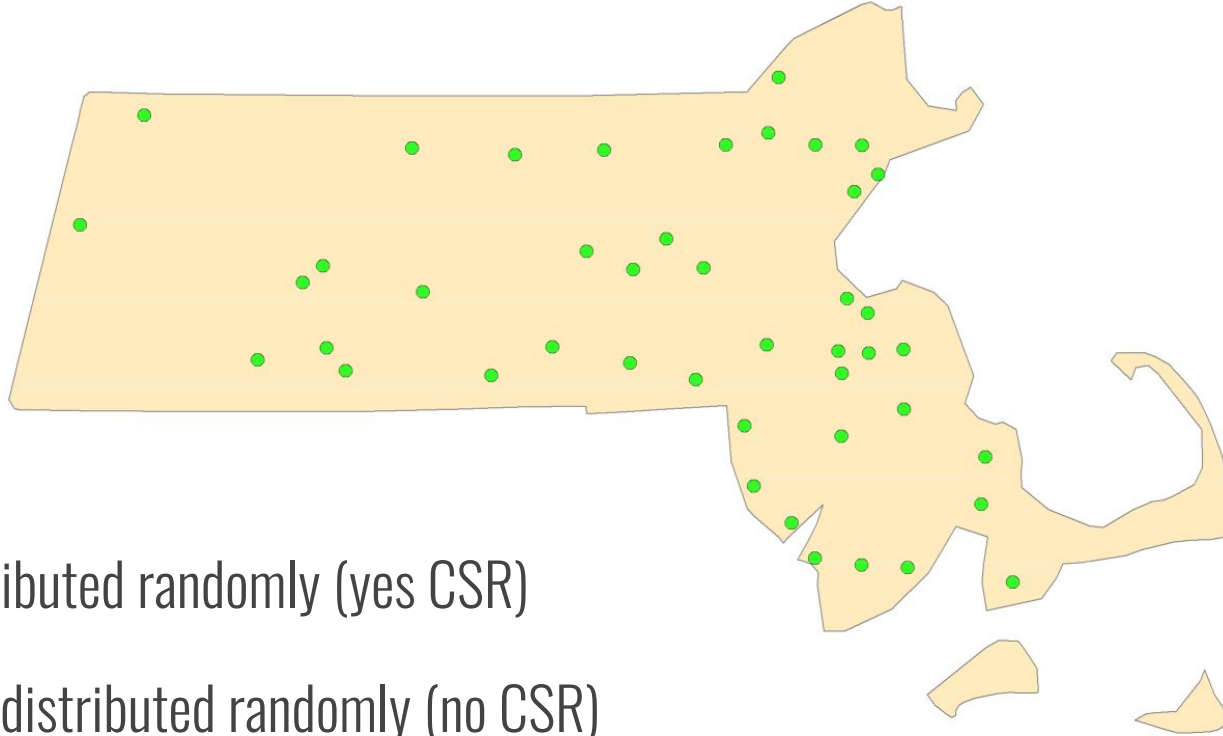
Compare observed point patterns to ones generated by an **independent random process (IRP)**, aka **complete spatial randomness (CSR)**.

CSR/IPR satisfy two conditions:

1. Any event has equal probability of being in any location, a 1st order effect.
2. The location of one event is independent of the location of another event, a 2nd order effect



Is this distribution of Walmarts in MA the result of CSR?



H_0 : Distributed randomly (yes CSR)

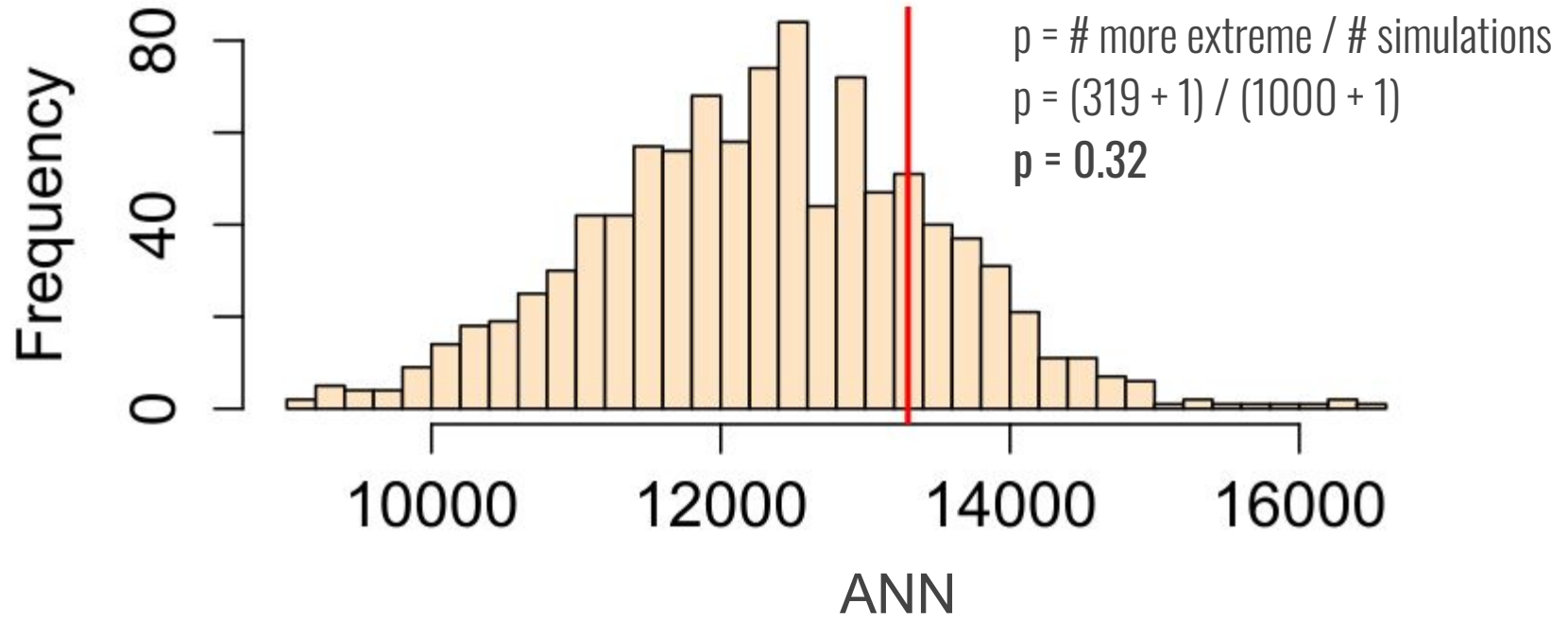
H_a : NOT distributed randomly (no CSR)

Hypothesis Testing: A Monte Carlo Test

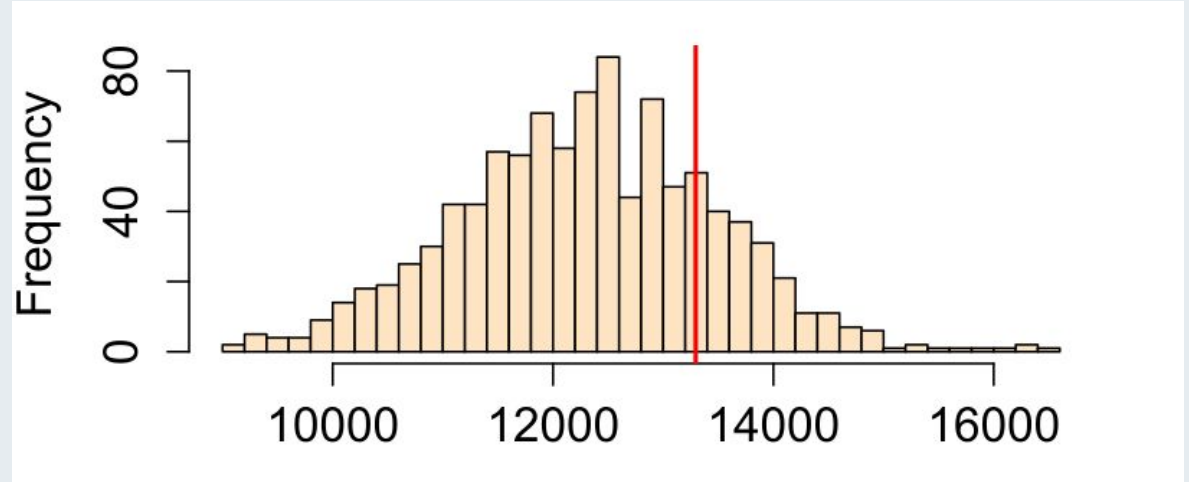
1. First, we postulate a process—**our null hypothesis, H_0** . For example, we hypothesize that the distribution of Walmart stores is consistent with a completely random process (CSR).
2. Next, we **simulate** many realizations of our postulated process and compute a statistic (e.g. ANN) for each realization.
3. Finally, we **compare our observed data to the patterns generated by our simulated processes** and assess (via a measure of probability) if our pattern is a likely realization of the hypothesized process.

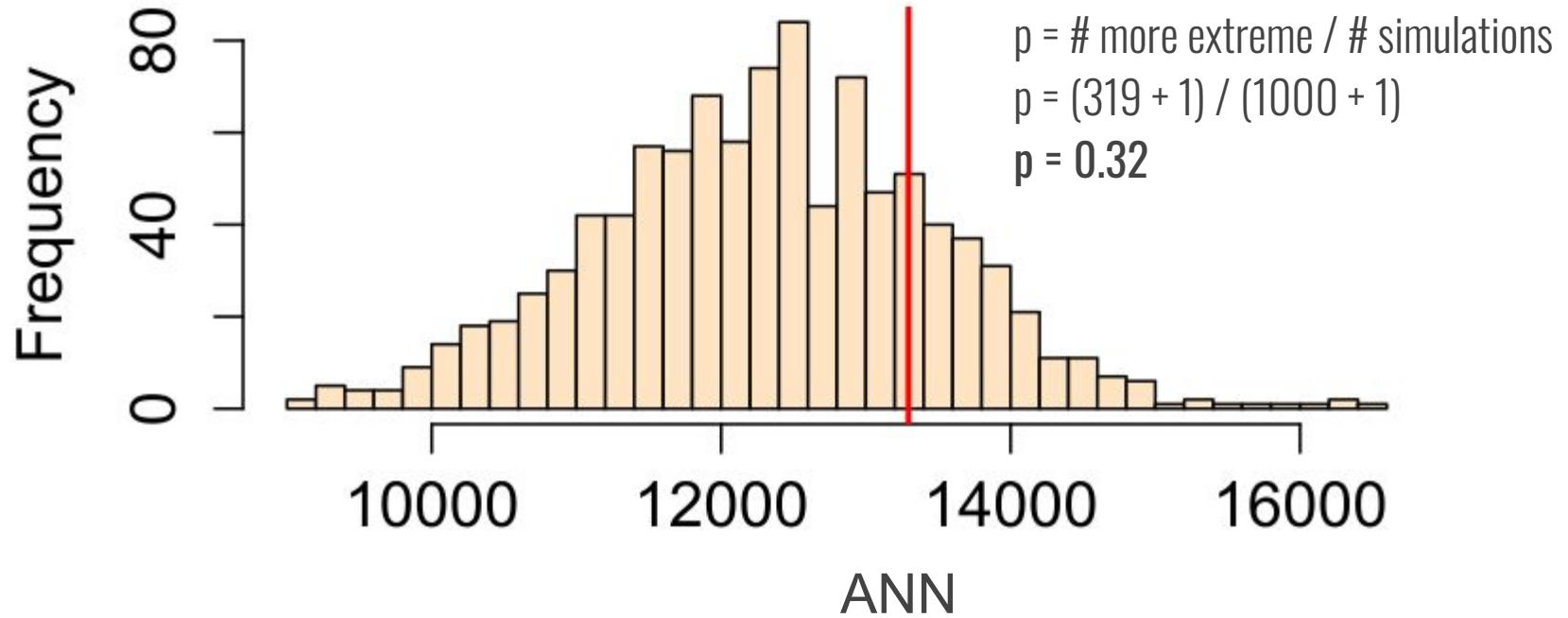


This is an example of bootstrapping!



Comprehension Check

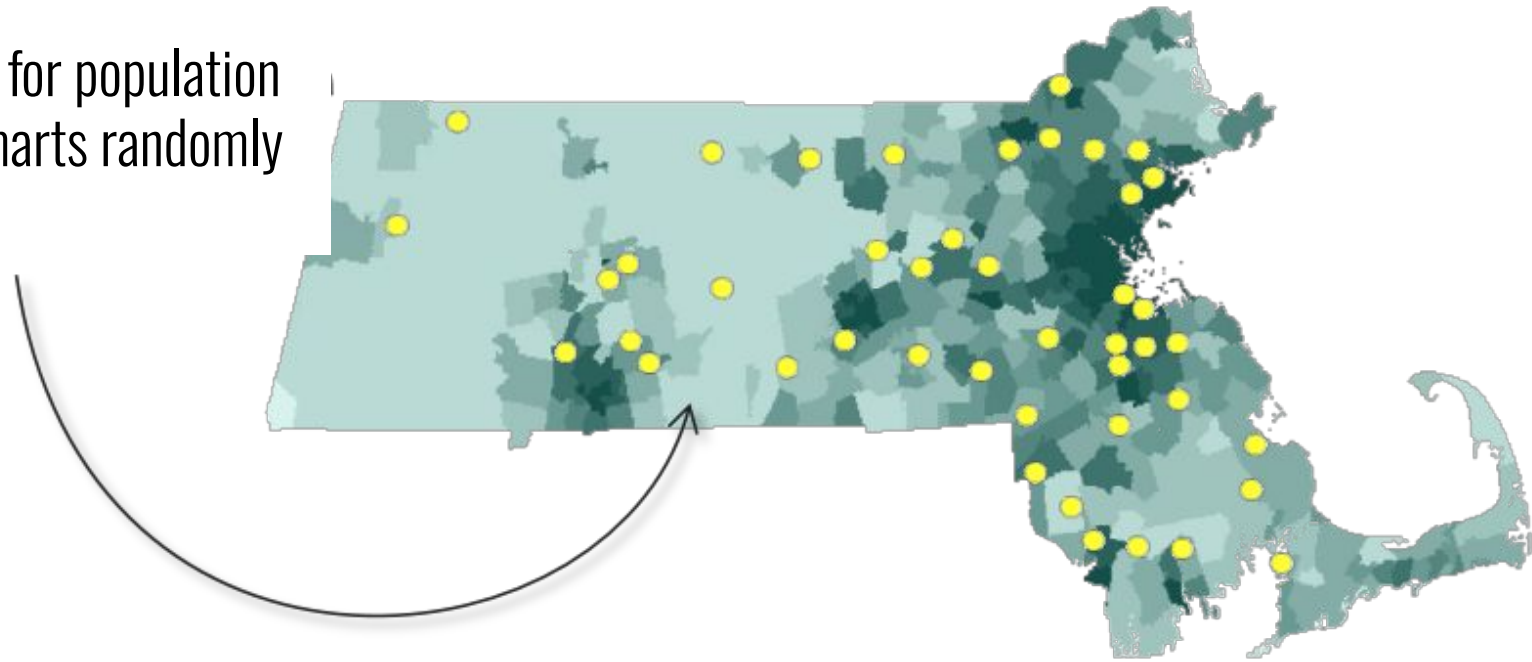




Fail to reject the null

Suggests that our results come from a CSR

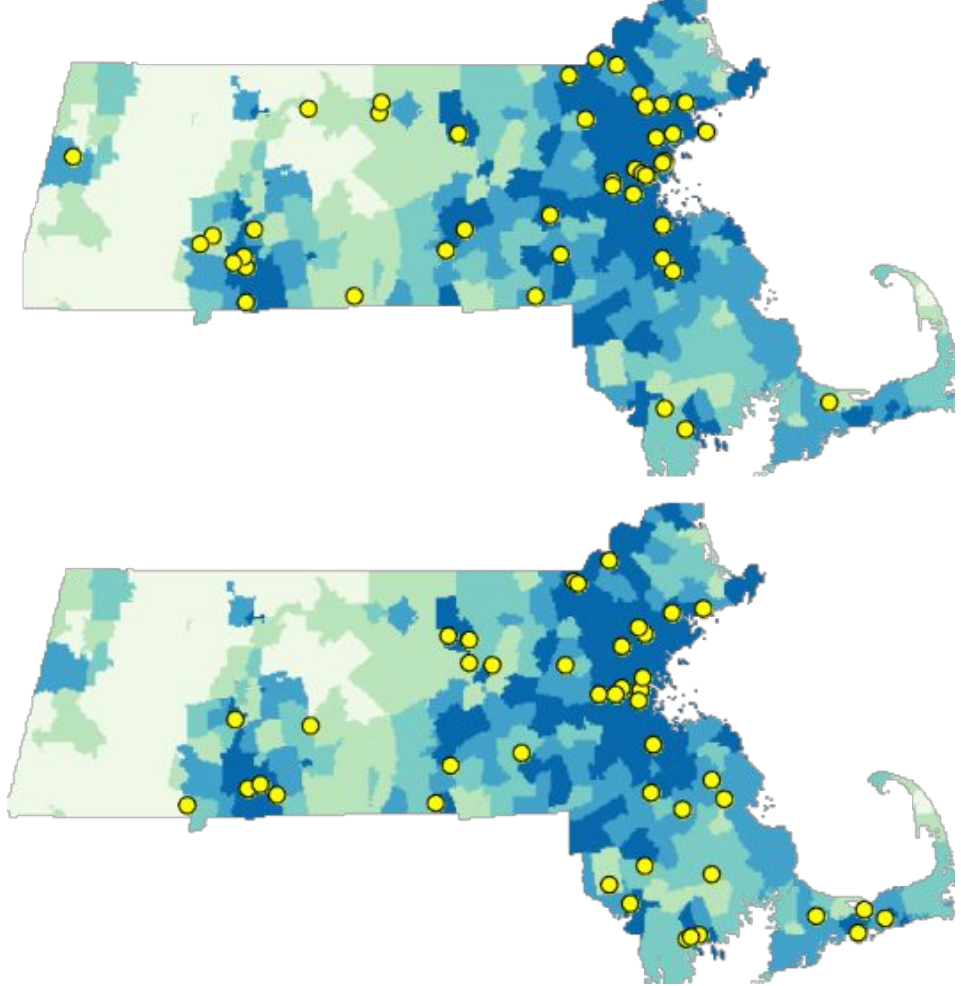
When controlling for population density, are Walmarts randomly distributed?

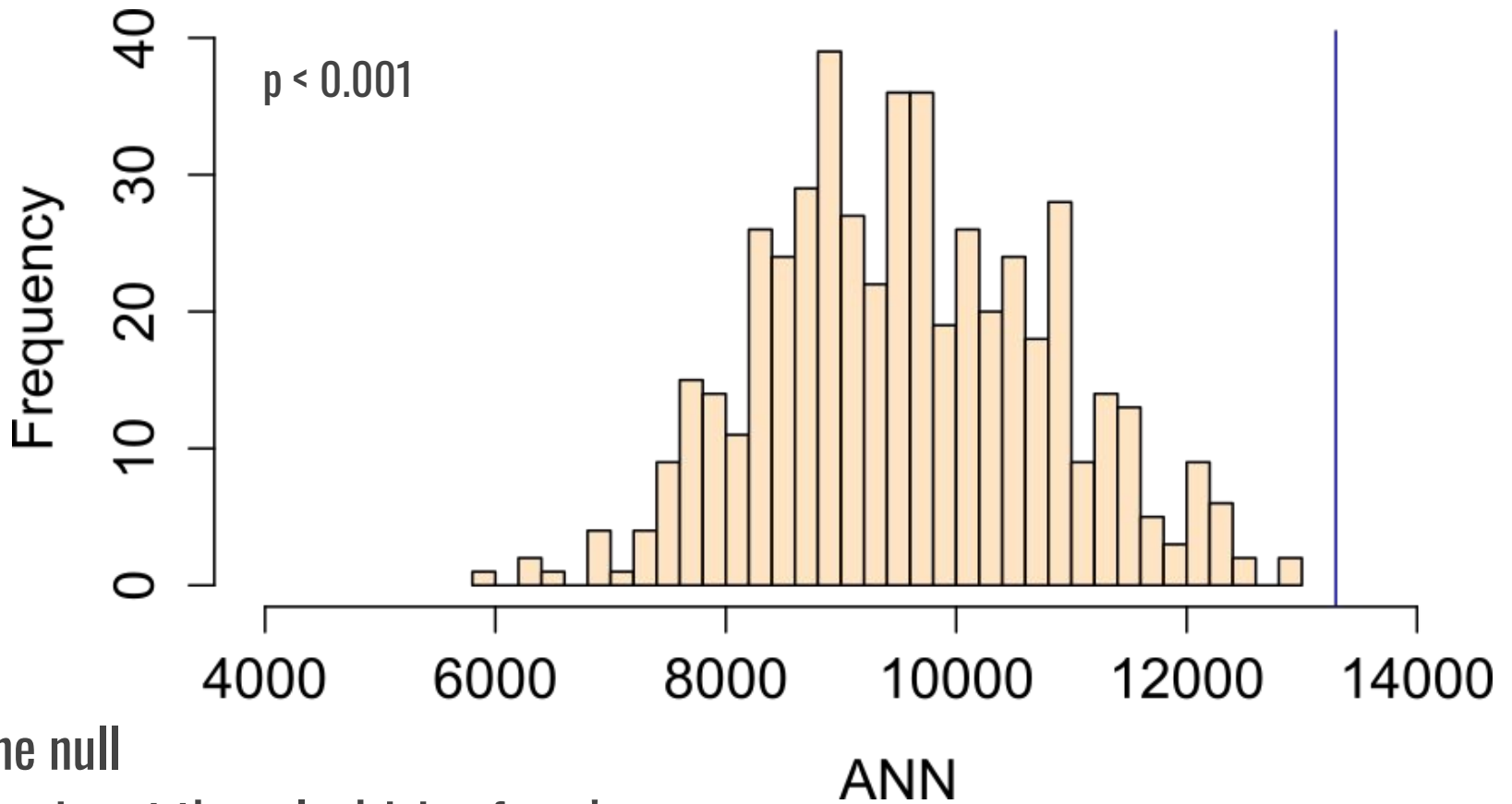


H_0 : Walmarts are distributed according to population density alone

H_a : Walmarts are *not* distributed based on population density alone

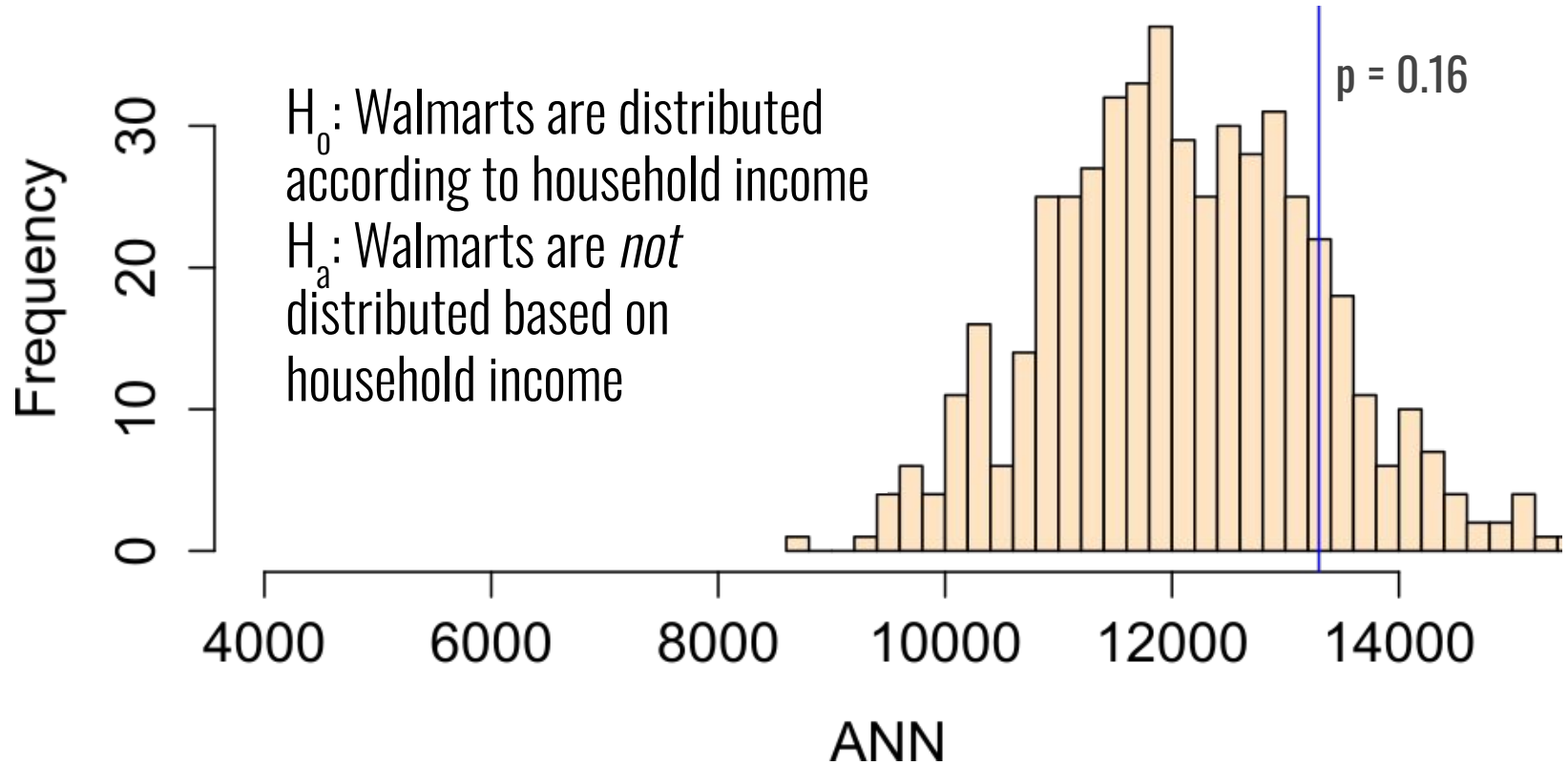
two randomly generated
point patterns using
population density as the
underlying process





Reject the null
Population is not the sole driving force!

Maybe median household income is the driving force...?



...Is it CSR or median household income?

hints at plausible scenarios, but doesn't tell us which one it is definitively.

Basic Geospatial Analysis: Summary

1. Considerations when visualizing spatial data important to conclusions drawn
 - a. values to plot?
 - b. map type?
 - c. color scale?
2. Traditional statistics fail with geospatial data:
 - a. Spatial autocorrelation
 - b. MAUP
 - c. Edge effects
 - d. Ecological fallacy
 - e. Nonuniformity of space
3. Analysis still possible
 - a. Global Point Density, Quadrat Density, Kernel Density
 - b. Poisson Point Process
 - c. Neighbor based methods: ANN; KNN
 - d. Comparison to a CRP (using simulation)